

BEYOND MEAN ATTENTION: DIVERSITY-AWARE, LAYER-WISE SCORING FOR KV CACHE EVICTION

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ABSTRACT

KV cache eviction methods such as SnapKV and PyramidKV rank tokens solely by *mean* attention over a small observation window. We study a unified score, $\mu_i + \lambda_1 \sigma_i + \lambda_2 \text{corr}(i, S)$, adding attention *dispersion* across window queries and *redundancy* relative to selected tokens. For $\lambda_2 < 0$, the score penalizes similarity to selected tokens as in maximal marginal relevance (MMR), without extra forward passes. To test whether this relevance–diversity balance should vary with depth, we compare fixed global coefficients with three-segment and quadratic profiles. Only these depth profiles are searched on a development split under a sinh reparameterization. On all 16 English LongBench datasets with Mistral-7B at a budget of 64 entries per layer, a single global diversification constant improves 13 of 16 datasets (macro +1.1); the gain holds at budget 32 and narrows at 128. Per-dataset search finds no detectable layer structure on most datasets; on passage retrieval it finds a large one: a mid-layer sign flip that rewards similarity and is worth +9.6 over the baseline at budget 64 and, without re-tuning, +13.2 over the global constant at budget 128. Ablations attribute the gain to the redundancy term; replaying every accepted search state on the held-out test set separates genuine structure from tuning noise.

Index Terms— KV cache compression, large language models, long-context inference, diversified selection

1. INTRODUCTION

Serving long-context large language models (LLMs) is dominated by the memory footprint of the key–value (KV) cache, which grows linearly with context length and often exceeds the model weights. Token eviction is a popular remedy: after prefill, each layer keeps only a small budget of B cached entries. State-of-the-art eviction methods such as SnapKV [1] and PyramidKV [2] score each historical token by the *mean* attention it receives from an observation window of the last few queries, and keep the top- B .

The mean, however, ignores two signals. First, window queries may agree or disagree about a token; the mean

discards this *dispersion*. Second, and more importantly, top-mean tokens are often near-duplicates of one another. At tight budgets, the cache then fills with *redundant* copies of one salient region while complementary evidence is evicted, the problem that maximal marginal relevance (MMR) addresses in information retrieval [3]. Inspired by mean–variance portfolio selection [4], we treat cache eviction as building a portfolio of tokens and study the unified score $\mu_i + \lambda_1 \sigma_i + \lambda_2 \text{corr}(i, S)$ with layer-wise coefficients.

Our contributions are threefold. (i) A diversified, layer-wise scoring framework for KV eviction that generalizes the attention-based token scoring of SnapKV/PyramidKV under a fixed budget allocation and costs no extra forward passes (Sec. 3). (ii) A comparison of a fixed global constant with two searched layer-wise parameterizations, Segments and Curve, under a random-split protocol in which the test set never influences any accepted search move (Sec. 3.3). (iii) A study on all 16 English LongBench datasets that shows a fixed global setting improving 13 of 16 datasets at $B=64$ without per-dataset search, and one large, transferable task-specific structure, together with ablations of the score terms, results for the global constant on a second model, and a *trajectory-replay* diagnostic that separates genuine structure from tuning noise (Sec. 4).

2. RELATED WORK

KV cache compression. StreamingLLM [5] keeps attention sinks plus a sliding window; H2O [6] and Scissorhands [7] evict by accumulated attention during decoding; FastGen [8] selects a per-head compression policy by profiling. Closest to us, SnapKV [1] scores prompt tokens by pooled observation-window attention at the end of prefill, and PyramidKV [2] adds a depth-decreasing budget schedule. These rank tokens *independently* by (a variant of) mean attention; our score contains theirs as its $\lambda_1=\lambda_2=0$ point. Redundancy-aware compression is a recent and active line: R-KV [9] combines attention importance with key-similarity redundancy for decoding-time compression; MixKV [10] balances importance and diversity per head, relative to the global key distribution, for vision-language caches; and the concurrent

TwinKV [11] adds a repair pass that swaps retained near-duplicate keys for evicted orphans. We differ on three axes: redundancy is *conditioned on the set already retained* through a greedy MMR rule, its weight is a searched *function of depth*, and the resulting profiles are tested for transfer across budgets.

Diversified subset selection. Selecting items that are individually relevant yet mutually complementary is classical: MMR [3] greedily penalizes similarity to the selected set, and determinantal point processes [12] model diversity probabilistically; mean–variance portfolio theory [4] makes the same trade-off for assets. We bring this view to KV eviction with a depth-dependent trade-off.

3. METHOD

3.1. Preliminaries

At the end of prefill, SnapKV/PyramidKV use an observation window W comprising the last $|W|=8$ queries. For a layer with cache budget B , token i is ranked by its mean attention, with the baselines’ max-pooling retained in the scoring pipeline:

$$\text{score}(i) = \mu_i = \frac{1}{|W|} \sum_{q \in W} A_{q,i}. \quad (1)$$

Here, $A_{q,i}$ denotes the attention received by token i from query q . The top- B tokens are retained. PyramidKV decreases budgets with depth while preserving the total budget of uniform allocation.

3.2. Diversity-Aware Scoring

Our *diversity-aware, layer-wise scoring for KV cache eviction* extends mean-attention ranking to capture query disagreement and redundancy among retained tokens. Inspired by mean–variance portfolio selection [4], we combine individual relevance with dispersion and set-dependent similarity. This analogy motivates the scoring components rather than a direct portfolio objective: attention statistics characterize each candidate, whereas key similarity relates it to the selected set.

Unified score. For a candidate i and the selected token set S , we define

$$\text{score}(i) = \mu_i + \lambda_1 \sigma_i + \lambda_2 \text{corr}(i, S), \quad (2)$$

where attention dispersion is

$$\sigma_i = \left(\frac{1}{|W| - 1} \sum_{q \in W} (A_{q,i} - \mu_i)^2 \right)^{1/2}, \quad (3)$$

and redundancy is measured by

$$\text{corr}(i, S) = \max\left(0, \max_{j \in S} \cos(k_i, k_j)\right), \quad \text{corr}(i, \emptyset) = 0. \quad (4)$$

Here, k_i is the cached key vector for token i . The sample standard deviation σ_i measures disagreement across window queries; $\text{corr}(i, S)$ measures the candidate’s greatest similarity to a retained token, with negative similarities clamped to zero. Thus, $\lambda_2 < 0$ gives an MMR-style penalty [3], favoring complementary evidence, whereas $\lambda_2 > 0$ rewards similarity. Because attention magnitudes vary substantially across heads but cosine similarity lies in $[-1, 1]$, we normalize the base score $s_i = \mu_i + \lambda_1 \sigma_i$ per head before greedy selection:

$$\tilde{s}_i = \frac{s_i}{\max_j s_j}, \quad S \leftarrow S \cup \left\{ \arg \max_{i \notin S} (\tilde{s}_i + \lambda_2 \text{corr}(i, S)) \right\}. \quad (5)$$

The denominator is clamped to a positive value, preserving the base-score ranking; $\tilde{s}_i \in [0, 1]$ when $\lambda_1 \geq 0$. The first token maximizes $\tilde{s}_i$, and subsequent admissions follow Eq. (5) until $|S| = B$. Similarity uses keys as cached, after rotary position embedding (RoPE), and therefore reflects both content and relative position. Setting $\lambda_1 = \lambda_2 = 0$ recovers the baseline ranking. The procedure uses existing attention and keys, requiring no additional forward passes; with candidates restricted to the top $4B$ tokens by base score, the B admissions cost $O(B^2 d)$ per head.

3.3. Layer-Wise Parameterization

The coefficients λ_1 and λ_2 control complementary aspects of selection and need not be shared across depth. Layer-dependent representations and PyramidKV’s nonuniform budget allocation motivate depth-varying coefficients, but limited development data favor compact profiles over independent tuning at every layer. Thus, we represent the coefficients with compact depth profiles.

Depth profiles. For $L=32$ layers, independently tuning both coefficients requires $2L$ parameters and risks overfitting 100 development questions. We instead compare three families for the depth profile $a(\ell)$: *Global*, a shared constant; *Segments*, separate constants for shallow, middle, and deep thirds; and *Curve*, a quadratic:

$$a(\ell) = c_0 + c_1 t + c_2 t^2, \quad t = \frac{\ell}{L-1}, \quad \lambda^{(\ell)} = \sinh(a(\ell)). \quad (6)$$

Layers are indexed by $\ell = 0, \dots, L-1$, and the mapping to $\lambda^{(\ell)}$ applies separately to λ_1 and λ_2 in every family. Segments and Curve each use three parameters per coefficient and include Global as a special case. For Curve, c_0 , c_1 , and c_2 control the latent profile’s height, tilt, and curvature, allowing at most one turning point. The sinh mapping is unbounded and sign-symmetric, approximately identity near zero, and satisfies $\cosh a \geq 1$. It also spans $\lambda \in [-3.6, 3.6]$ approximately over $a \in [-2, 2]$.

Search protocol. For each dataset, a fixed-seed random split yields a development set of 100 questions (250 for lcc

and repobench-p) and a held-out test set of the remaining questions. Only development scores determine accepted search moves; test scores serve reporting and trajectory replay (Sec. 4.3). Because λ_1 reshapes base scores and λ_2 changes the resulting greedy selections, we optimize them jointly by first-improvement coordinate search over the six parameters of Segments or Curve; Global keeps the fixed setting $\lambda_1=0$, $\lambda_2=-0.15$ and is not searched. This setting is also the warm start of every search. A pilot comparing $\lambda_2 \in \{-0.15, -0.3, -0.6\}$ on qasper, hotpotqa, 2wikimqa, and passage retrieval at $B=64$ fixed this constant, which is reused across all 16 datasets, budgets, and both models. Each search allows 14 evaluations, perturbing coordinates by ± 0.3 in a -space. A move is accepted only when its development-score improvement exceeds τ . Near zero, a 0.3 perturbation in a corresponds approximately to a 0.3 change in λ ; for the redundancy coefficient, this can shift a candidate’s normalized score by about 0.3 because $\cos \in [-1, 1]$. Larger steps help expose effects within the limited evaluation budget, although their effective magnitude varies with a . With per-question score standard deviation ≈ 35 , the development standard error is ≈ 3.5 points. We use an acceptance tolerance of $\tau=0.1$ points; this is not a statistical significance threshold.

4. EXPERIMENTS

4.1. Experimental Setup

Datasets We use LongBench [13] to assess the performance of our method on tasks involving long-context inputs. Its 16 English datasets cover single- and multi-document question answering, summarization, few-shot learning, synthetic retrieval and counting, and code completion.

Experiment settings We use Mistral-7B-Instruct-v0.2 [14] (fp16, SDPA) in the PyramidKV implementation with a budget of $B=64$ entries per layer ($\approx 1\%$ of a typical prompt) unless stated otherwise, and all 16 English LongBench [13] datasets. Test scores use the official LongBench scorer, which truncates predictions to their first line on trec, triviaqa and samsum; search-time dev scores use raw outputs. All comparisons share the same total cache budget. The full study took ≈ 2500 search trials and ≈ 260 GPU-hours on RTX 4090s.

4.2. Main results

Table 1 demonstrates the effectiveness of our diversity-aware, layer-wise scoring for KV cache eviction at $B=64$. With fixed coefficients $\lambda_1=0$ and $\lambda_2=-0.15$, Global improves over PyramidKV on 13 of 16 datasets, increasing the macro score from 33.31 to 34.44 (+1.13 points). Dataset-specific search with Segments reaches 35.15, whereas Curve obtains 34.33, indicating that greater profile flexibility does not consistently improve performance. The strongest task-specific benefit occurs on passage retrieval: Segments achieves 70.17, exceeding PyramidKV by 9.57 points and Global by 3.36 points.

Dataset	PyramidKV	Global	Ours	
			Seg.	Curve
qasper	23.73	25.12	25.12	25.12
hotpotqa	33.67	36.94	36.94	36.17
2wikimqa	23.52	23.33	23.76	23.12
passage-retrieval	60.60	66.81	70.17	65.87
musique	13.13	11.88	12.07	12.00
narrativeqa	18.10	18.42	18.85	18.42
multifieldqa	32.14	33.93	35.05	33.93
trec	55.00	56.00	61.00	56.00
triviaqa	84.53	85.21	85.33	84.49
samsum	33.50	35.23	35.48	35.23
passage-count	4.18	5.61	5.61	5.61
lcc	49.88	49.51	49.51	49.51
repobench-p	41.51	41.96	41.96	41.96
gov-report	18.42	19.08	19.09	19.08
qmsum	20.90	20.97	21.38	21.65
multi-news	20.17	21.04	21.04	21.04
Macro	33.31	34.44	35.15	34.33
Wins	2	0	13	1

Table 1. Test scores (Mistral-7B, budget 64, official LongBench scorer). Bold: row best. Wins: number of datasets on which the column is the row best (ties credited to Seg.).

Method	$B=32$	$B=64$	$B=128$
PyramidKV	30.48	33.31	35.79
Global	31.21	34.44	35.99
Seg.	31.49	35.15	36.39

Table 2. Budget sweep on test sets (Mistral-7B). We report the macro average performance; Seg. profiles are deployed at $B=64$ and transferred unchanged.

These results support fixed diversification as an effective default, with additional benefits from depth-dependent scoring on selected tasks.

Table 2 varies the budget with the profiles deployed at $B=64$ transferred unchanged. Global improves the macro score over PyramidKV by 0.73, 1.13, and 0.20 points at $B=32$, 64, and 128, and Seg. by 1.01, 1.84, and 0.60 points, respectively. The passage-retrieval profile transfers unchanged (Fig. 1c): it scores 63.03 at $B=32$ and 81.70 at $B=128$, 13.40 above PyramidKV and 13.16 above Global at the latter budget.

4.3. Ablation studies and further analysis

Visualization of passage retrieval results. Fig. 1 shows the passage-retrieval profile (a), its search trajectory (b), and its budget transfer (c). The profile keeps $\lambda_2 < 0$ in shallow layers but turns it positive in the middle and deep layers. In the replay, the two accepted λ_1 updates do not improve the test score, whereas the λ_2 sign flip yields a large gain, and the

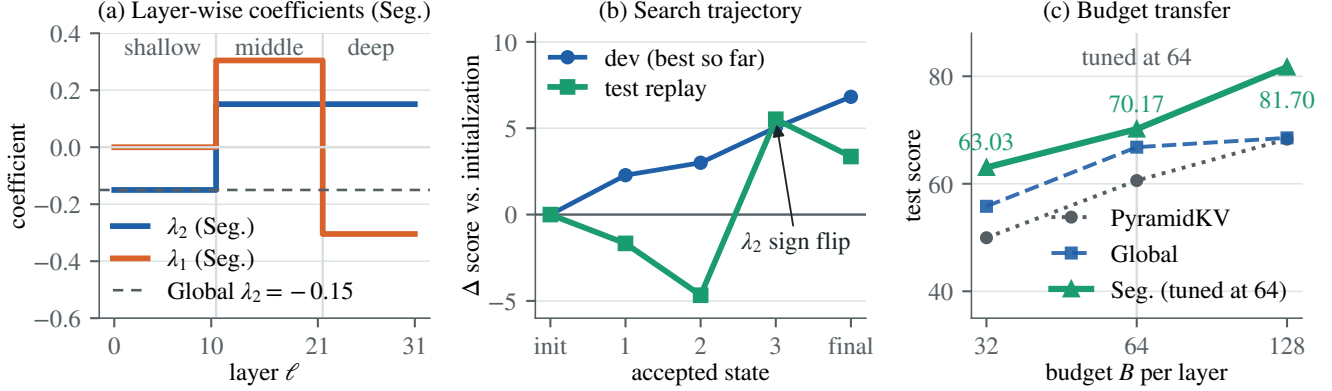


Fig. 1. Passage retrieval, Mistral-7B. (a) Layer-wise coefficients of the Seg. profile searched at $B=64$; dashed line: Global $\lambda_2 = -0.15$; vertical lines: segment boundaries. (b) Development best-so-far and test replay of every accepted state of the search at $B=64$ (14 evaluations, $\tau=0.1$), as change from the initialization; only development scores determine acceptance. (c) Test score vs. budget: the Seg. profile tuned at $B=64$ is applied unchanged at $B=32$ and 128.

Scoring rule	Macro	Wins
Mean (baseline)	33.31	—
Mean & dispersion	33.50	8/16
Mean & Corr	34.57	13/16
dispersion & Corr (fixed λ)	33.16	9/16
All three terms (ours, 3 segments)	35.15	14/16
All three terms, 2 segments	34.75	14/16
All three terms, 8 segments	35.03	14/16

Table 3. Ablation studies of our framework (Mistral-7B, budget 64), with the Seg. strategy unless noted. We report the macro average scores (Macro) and the number of datasets improved over the baseline (Wins).

profile tuned at $B=64$ exceeds both PyramidKV and Global at every budget.

Ablation study of our framework. Table 3 removes one score term at a time and varies the depth granularity. Mean attention is the backbone: a fixed rule without it (dispersion & Corr, $\lambda_1=0.3$, $\lambda_2=-0.15$, no search) scores 33.16, below the 33.31 baseline, and drops passage retrieval to 57.41. The redundancy term carries most of the gain: searching λ_2 alone (Mean & Corr) reaches 34.57 and improves 13 datasets, whereas searching λ_1 alone (Mean & dispersion) reaches 33.50 and improves 8. Searching both reaches 35.15; on passage retrieval the joint search scores 70.17 against 66.62 for λ_2 alone in this round. Three segments score highest (35.15); eight segments (35.03; 16 parameters, 34 evaluations) and two (34.75) stay within the noise floor of it, so the segment count is not a sensitive choice.

Evaluation on other LLM backbones. To check that the fixed rule transfers across models, we apply Global ($\lambda_1=0$, $\lambda_2=-0.15$, no re-tuning) to Llama-3.1-8B-Instruct at $B=64$. gasper improves by +5.8 (24.4 $\rightarrow$ 30.1) and hotpotqa by

+2.0 (51.1 $\rightarrow$ 53.1), while the other datasets stay within the noise floor; passage retrieval is already at 99.0 on this model, leaving limited headroom for a depth profile. Per-dataset search on this model is left for future work.

Discussion on efficiency and search cost. The greedy admission runs once per prompt at the end of prefill and adds no forward passes. On 12 HotpotQA prompts (mean 15.6k tokens, RTX 4090) selection takes 222 ms per prompt versus 32 ms for top- k , increasing end-to-end latency for prefill plus 32-token generation by about 14% (2.52 s $\rightarrow$ 2.87 s); the overhead comes from performing the B admissions sequentially in each layer. Global needs no per-dataset search. A three-segment profile costs 14 development evaluations per dataset, about 1–3 GPU-hours on one RTX 4090, and transfers across budgets without re-tuning (Table 2).

Limitations. Scores are single-run on one benchmark suite with test sets of 50–250 questions per dataset, so differences of a few points on a single dataset should be interpreted with caution; greedy search is path-dependent, and per-dataset profiles need labeled development data.

5. CONCLUSION

Diversified, layer-wise scoring generalizes mean-attention token scoring for KV eviction without additional forward passes. On Mistral-7B a single global diversification constant is a robust default at small cache budgets, and one task, passage retrieval, rewards a layer-wise profile with a large gain that survives changes of budget. Ablations place the benefit in the redundancy term computed on cached keys, and a trajectory-replay diagnostic makes such findings verifiable. Next steps are instance-adaptive coefficients.

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7. COMPLIANCE WITH ETHICAL STANDARDS

This computational study used the publicly available LongBench benchmark [13]. No new data were collected, and no human or animal subjects were involved; therefore, ethical approval was not required.

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